

**AMENDMENT NO. 1 JULY 2001
TO
IS 7098 (Part 3) : 1993 CROSS-LINKED
POLYETHYLENE INSULATED THERMOPLASTIC
SHEATHED CABLES
PART 3 FOR WORKING VOLTAGES FROM 66 kV UPTO AND
INCLUDING 220 kV**

(Page 1, clause 1.1, line 4) — Substitute following for the existing line:
'without metallic sheath and thermoplastic'.

(Page 1, clause 6) — Insert the following after (c):
'd) Copper woven fabric tape.'

(Page 2, clause 7.2) — Substitute the following for the existing:

7.2 Metallic Sheath

7.2.1 The metallic sheath shall be either lead alloy sheath or aluminium sheath.

7.2.2 Composition of lead alloy 'E' sheath shall be as per Table 3.

7.2.3 Composition of aluminium for sheathing

Aluminium for sheathing shall have minimum 99.6% by weight of aluminium.'

(Page 2, Table 3, clause reference) — Substitute '7.2.2' for '7.2'.

(Page 2, Table 3, col 3) — Substitute '0.35' for '0.34'.

(Page 2, Table 3, col 10 and 11) — Substitute the following for the existing:

Total of other Elements	Lead
<i>Max</i>	<i>Min</i>
(10)	(11)
0.01	Remainder'

(Page 3, clause 10.1, line 1) — Substitute 'shall' for 'may'.

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(Page 3, clause 14.3.1) — Substitute the following for the existing clause:

'14.3.1 The metallic part of the screening shall consist of either copper wires/copper tape(s) helically applied over non-metallic screening or an extruded metallic sheath of aluminium or lead alloy. The metallic sheath applied over the non-metallic part of the insulation screen shall form the metallic part of the screening.

NOTE — For enhancing short-circuit current carrying capacity of screen, additional copper wires/tape (s) can be applied over/under Lead alloy sheath with a suitable separator.'

(Page 4, clause 14.3.3, Note, line 3) — Substitute 'may' for 'shall'.

(Page 4, clause 15.1) — Substitute the following for the existing clause:

'15.1 The moisture barrier shall be provided as given in 15.1.1 or 15.1.2 or 15.1.3.

15.1.1 Water blocking tape/powder shall be suitably applied and laminate tape shall be applied longitudinally with overlap and sealed.

15.1.2 Metallic sheath of lead alloy 'E' as per 7.2 shall be applied by extrusion. thickness of sheath shall not fall below the values specified in Table 6.

15.1.3 Alternative to above, moisture barriers shall be provided by Aluminium sheath either smooth or corrugated.'

(Page 4, clause 15.2.1) — Substitute the following for the existing clause:

15.2.1 Metallic sheath shall consist of either extruded lead alloy 'E' Sheath as per 7.2.2 or Aluminium Sheath smooth or corrugated as per 7.2.3.'

(Page 4, clause 15.2.2) — Substitute the following for the existing clause:

'15.2.2 Thickness of lead alloy sheath shall not fall below the value specified in Table 6A. Thickness of aluminium sheath shall not fall below the value specified in Table 6B.

NOTE — Aluminium sheath can be either smooth or corrugated. From flexibility and thermal expansion point of view corrugated aluminium sheath is recommended. Smooth aluminium sheath is not recommended for cables of calculated diameter under metallic sheath exceeding 50 mm.'

(Page 4, Table 6A, col 2, row 3) — Substitute '50' for '45'.

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(Page 4, Table 6, Title) — Substitute the following for the existing title:
'Table 6A Thickness of Lead Alloy Sheath'.

(Page 4, Table 6) — Insert the following new Table 6B after Table 6:

Table 6B Thickness of Aluminium Sheath

Calculated Diameter under Metallic Sheath mm		Minimum Thickness of Metallic Sheath mm	
Over	Up to and including	Smooth	Corrugated
(1)	(2)	(3)	(4)
—	40	1.9	0.6
40	45	2.2	0.6
45	50	2.4	0.6
50	55	—	0.6
55	65	—	0.8
65	80	—	0.8
80	105	—	0.9
105	120	—	1.0

(Page 4, clause 17.2, line 1) — Substitute 'nominal diameter' for 'dimensions'.

(Page 4, Table 8, Title) — Substitute 'Diameter' for 'Dimensions'.

[Page 5, clause 18.1(a)] — Substitute the following for the existing:

'a) over the moisture barrier or metallic screen for unarmoured cables, and'

(Page 5, clause 18.4) — Substitute the following for the existing clause:

'18.4 The outer sheath shall be coated with a suitable conducting material when required.'

[Page 5, clause 19.1(b)(i), col 2] — Substitute 'Table 5 and 20.16' for 'Tables 5, 19, 16'.

[Page 6, clause 19.1(f)(i), col 1] — Substitute 'Diameter of armour wire' for 'Dimensions'.

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[Page 6, clause **19.1(g)(3)** (i), (ii) and (iii)] — Substitute the following for the existing:

'3)	PE sheath	Nominal value	Part 32
I)	Carbon black content	2 5 ± 0.5%	
II)	Tensile strength	8 3	Part 7
III)	Elongation at break		
	Before ageing	8 3	Part 7
	After ageing	8 3	Part 7
IV)	Hot deformation	8 3	Part 15'

[Page 6, clause **19.1 (j)**] — Substitute the following for the existing:

???	Water tightness test		
I)	Longitudinal water lightness test	20 8 1 and 20 8 3	Annex D
II)	Radial water lightness test	Under consideration	Under consideration'

[Page 6, clause **19.1(k)(ii)**, col 2]— Substitute '20 9.1' for 'Table 2 and IS 5831 : 1984'.

(Page 6, clause **19.1(k)** (iii), col 2] — Substitute '20.9.1 and Annex C' for '20.2'.

[Page 6, clause **19.1(k)** (iii), col 3] — Substitute '20.2.2 and Annex C' for '20.2'.

(Page 6, clause **19.1, Note 1**) — Substitute 'm to r' for 'n to q'.

(Page 6, clause **19.1, Note 2, line 1**) — Substitute 'm to r' for 'n to r'.

(Page 7, clause **19.1.1, line 1**) — Substitute 'm to r' for 'n to r'.

(Page 7, clause **19.1 2, line 1**) — Substitute 'n and q' for 'p and q'.

(Page 7, clause **19.4**) — Substitute the following for the existing clause:

'19.4 Optional Test

a) Cold Impact Test

b) Voltage Test (spark test on outer sheath without semi-conducting coating)

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A spark test shall be carried out by earthing the metallic sheath or concentric wires or tapes for an ac test and connecting them to the negative pole in the case of a dc supply. The voltage shall be 6 kV ac per millimeter nominal thickness or 9 kV dc per millimeter nominal thickness of the extruded oversheath subject to maximum values of 15 kV and 25 kV respectively. The dwell time of the cable in the region of test shall have sufficient duration to detect any defect.'

(Page 7, clause **19.5**, para 3, line 6) — Substitute 'same' for 'ssme'.

(Page 7, clause **19.5**) — Insert the following Note at the end of the clause:

'NOTE — The declared nominal value of the insulation thickness in context with 19.5 shall mean the value declared by the manufacturer to the purchaser.'

(Page 7, clause **20.1.1**, para 4, line 5) — Substitute '1.27 mm' for '0.127 mm'.

[Page 7, clause **20.1.1**, para 6, Sl No.(i), (ii) and (iii)] — Substitute '50 μm ' for '5 μm '.

(Page 7, clause **20.1.2**) — Delete this clause.

(Page 8, clause **20.4**, Note, line 2) — Substitute '0.017 86' for '0.178 6'.

(Page 8, clause **20.5**, last line) — Substitute 'either in Table 6A or Table 6B' for 'in Table 6'.

(Page 8, clause **20.6**) — Substitute the following for the existing clause:

'20.6 Diameter of Armour

The diameter of non-magnetic armour wires shall not be less than the specified nominal values given in Table 8.'

(Page 8, clause **20.8.1**, heading) — Substitute the following title for the existing:

'Water Tightness Test (Not Applicable for corrugated Aluminium Sheathed Cables)'.

(Page 8, clause **20.9.1**, para 2, line 2) — Substitute 'Annex C' for 'Annex B'.

(Page 8, clause **20.10**, para 2, lines 2 and 4) — Substitute 'Uo' for 'the rated voltage'.

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[page 9, clause **20.11**,(i)] — Substitute 'metallic sheath' for 'lead sheath' and 'laminated tape' for 'metal foil'.

(Page 9, clause **20.12**, lines 2 and 4) — Substitute '*U_o*' for 'rated voltage'.

(Page 9, clause **20.13**, para 2, line 2) — Substitute '*U_o*' for 'rated voltage'.

(Page 9, clause **20.15**) — Insert the following at the end of the clause:

'The impulse wave form shall have a virtual front time between 1 μ s and 5 μ s and a nominal time to half the peak value between 40 μ s and 60 μ s. In other respects, the impulse pulse shall be in accordance with IS 2071.'

(Page 10, clause **20.17**, lines 3 and 6) — Substitute '*U_o*' for 'rated voltage'.

(Page 11, Table 10, col 4, row 6) — Substitute '50.5' for '50.3'.

(Page 11, clause **B-3.4**, Note, line 2) — Substitute '**B-3.3** and **B-3.4**' for '**A-3.3** and **A-3.4**'.

(Page 11, clause **B-3.5**) — Substitute '*D_{sc}*' for '*Sec*'.

(Page 13, clause **D-3**, line 4) — Insert '3' before the words 'heating cycles'.

(Page 13, clause **D-3**, line 5) — Substitute '16 h' for '15 h'.

(Page 13, clause **D-3**, line 11) — Substitute 'last' for 'first'.

(Page 13, Fig. 2) — Insert the following above the caption of the figure:

'The minimum inner diameter of the water tube shall be 10 mm'.

(ETD 9)